For this enhancement, I’m using the same artifact as for the first one. I selected my Android inventory management application.

I added a search feature that uses linear search to find inventory items by name within an ArrayList. I also added four sorting options that allow users to sort by name or quantity in ascending or descending order. Instead of relying on Java's sorting method, I created an InventoryAlgorithms class that implements merge sort. Linear search has a time complexity of O(n), while merge sort has a time complexity of O(n log n), making it a more efficient choice as the inventory grows.

This enhancement supports course outcomes three and four by demonstrating my ability to select and implement appropriate algorithms and data structures. I used an ArrayList to manage inventory objects and implemented searching and sorting algorithms to solve a practical problem within the app. I also separated these algorithms from the user interface, again improving the organization.

I used this enhancement to help myself understand the difference between using an existing algorithm and implementing one myself. Replacing Java's built-in sorting with merge sort required me to understand how the algorithm divides and merges data (GeeksforGeeks, 2026). Implementing linear search also gave me a practical example of O(n) complexity.

Reference

GeeksforGeeks. (2026, August 6). Merge sort. https://www.geeksforgeeks.org/dsa/merge-sort/